Revision No. CE/O-1124/00

# Sinetrac®

Power Products

## Chime Electronics

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Defence Approved

## PRODUCTS

- AUTOMATIC VOLTAGE STABILIZERS (SINGLE PHASE)
- SERVO VOLTAGE STABILIZERS (MICROPROCESSOR BASED)
- ISOLATION TRANSFORMERS
- ONLINE U.P.S. SYSTEMS
- STREET / LIGHTING TIMERS - UTILITY GADGETS
- WATER LEVEL INDICATORS - SENSORS
- PHASE SEQUENCE CORRECTORS (AUTO) / AUTOMATION PROTECTORS
- BATTERY CHARGERS(CCCV TYPE) / SWITCH TRIPPING /RECTIFIERS
- SOLAR U.P.S. HYBRID ADVANCED / PCU (DSP SINEWAVE)

# TECHNICAL SPECIFICATIONS

** Due to continuous upgradation in technology & customer needs the above specifications are subject to change.*

# Automatic Line Voltage Stabilizers

Automatic Digital Wall Mounting Stabilizer

Sinetrac®

## WALL MOUNTING Models

Sinetrac's Automatic Line Voltage Stabilizers are different from the one's normally available in local markets. A stabilizers is installed for the safety/smooth working of costly equipments which could adversely effect, hence right choice of cost & energy effecient stabilizers is important. A cheap stabilizer might cost less initially but because of low efficiency bores a hole in pocket due to heavy electricity bills.

## Exclusive features of sinetrac Automatic line Voltage stabilizers

- All stabilizers carry transformers made in best core giving high efficiency, consuming lowest possible electricity.
- All standard company components like Philips, Keltron, Bel, CDIL used in circuitary giving more reliability and life span.
- Wide ranging models to cater to voltages as low/high as 100 V–280V i.e. LR models 140V–280V i.e. UR models, 160V–260V i.e. UU models.
- Time delay high/low cut off facility reducing the risk of damage on out of range high & low voltages. Time delay by pass for new models A.C.'s & upto 180 seconds time delay factory setable for gases to settle in compressor, enhancing the compressor life to almost double.
- Integrated circuits (I.C.'s) used for precise and sharp sensing.
- Analog, metering off facility & Digital display, in select models.

### Model : WMD UUT

- Wall mounting : WM
- Metering : Digital – D
- Input voltage correction range : 160 v – 260 volts
- Out of correction range protection : High voltage cut off 270V (±) 5 volts
- Out of correction range protection : Low voltage cut off 160V (±) 10 volts
- Output voltage range : 225V (±) 11%
- Time Delay : Normal 30 sec. as inbuilt in new A.C's or 120 sec. (±) 3o sec.
- Capacity : 4kVA & 5kVA for A.C's

### Model : WMA UUT

- Wall mounting : WM
- Metering : Analog – A
- Input voltage correction range : 160 v – 260 volts
- Out of correction range protection : High voltage cut off 270V (±) 5 volts
- Out of correction range protection : Low voltage cut off 160V (±) 10 volts
- Output voltage range : 225V (±) 11%
- Time Delay : Normal 30 sec. as inbuilt in new A.C's or 120 sec. (±) 3o sec.
- Capacity : 4kVA & 5kVA for A.C's

### Model : WMD UT

- Wall mounting : WM
- Metering : Digital – D
- Input voltage correction range : 140 v – 280 volts
- Out of correction range protection : High voltage cut off 290V (±) 5 volts
- Out of correction range protection : Low voltage cut off 130V (±) 10 volts
- Output voltage range : 225V (±) 11%
- Time Delay : Normal 30 sec. as inbuilt in new A.C's or 120 sec. (±) 3o sec.
- Capacity : 4kVA & 5kVA for A.C's

### Model : WMA UT

- Wall mounting : WM
- Metering : Analog – A
- Input voltage correction range : 140 v – 280 volts
- Out of correction range protection : High voltage cut off 290V (±) 5 volts
- Out of correction range protection : Low voltage cut off 130V (±) 10 volts
- Output voltage range : 225V (±) 11%
- Time Delay : Normal 30 sec. as inbuilt in new A.C's or 120 sec. (±) 3o sec.
- Capacity : 4kVA & 5kVA for A.C's

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# SERVO VOLTAGE STABILIZERS

## MICROPROSSESOR BASED

Three Phase Oil Cooled Servo

Sinetrac®

Single Phase Air Cooled Servo

| | | |
|---|---|---|
| Capacity | : | 1kVA to 1000 kVA |
| Type | : | Unbalanced i.e. separate correction in each phase |
| Input Working Range (Standard Availability): | | Single Ph.:173-277V / 160-260V / 140-280V / 120-280V<br>Three Ph. :300-480V / 277-450V / 243-485V / 208-485V<br>{or to order} |
| Output Voltage | : | 220V / 230V /240V Setable Single Phase<br>380V / 400V /415V Setable Three Phase |
| Response Time | : | 5 milli sec. |
| Speed of Correction | : | 8V to 35V per phase i.e. 15V to 70V depending upon capacity & range stepless using variable transformer |
| Output Voltage Regulation | : | 1% to 2.5% (depending on output set) |
| Efficiency | : | **98%** (at rated load) (**Greater than 95%**) |
| No Load Losses | | Less than 1% (as Best Quality lamination used in transformer), True value return within a calendar year due to high energy savings |
| Effect of Power Factor on O/P Voltage | : | NIL |
| Waveform Distortion on O/P Voltage | : | NIL |
| **Winding Wires** & **Connecting Cables** | : | Pure electrolyte grade **copper not aluminium** |
| Input Frequency Variation | : | |
| 45 to 55 Hz, effect on output voltage | | NIL |
| By-Pass Arrangement | : | Optional* |
| Additional Facilities | : | Alarm Annunciator & manual operation* |
| Type of Cooling | : | Air Cooled upto 30KVA & oil cooled above that or as desired & dependent on input range |
| Enclosure & Mounting | : | **Powder coated** & On wheels in higher capacities |

**METERING: ANALOG / DIGITAL.* (Digital Display in Microprocessor based)**

- Input /output voltage phase to phase & neutral to phase.
- Ampere meter for Output Current per phase (optional).

**PROTECTIONS:**

- Under Voltage Protections i.e. cut off
- Over Voltage Protection i.e. cut off
- MCB or MCCB for Load Protection on Input side or output side.
- Single Phasing Protection i.e. shut down (optional)
- Phase Reversal Protection i.e. shut down (optional)

**** Due to continuous upgradation specifications may change without prior notice.***

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Sinetrac®

# ISOLATION TRANSFORMERS

Line transients and spikes, are the main factors, causing wrong operation, erratic performance and inaccurate recording of sensitive and Sophisticated Electronic Systems. To over come this problem, "SINETRAC" has come up with the answer of giving Isolation Transformer (in Conjunction with SINETRAC Automatic Voltage Stabilizers) to render clean Power for all sorts of sensitive electronic equipment.
"SINETRAC" Isolation Transformer ensures clean power supply to sensitive electronic equipments in addition to isolating the load from the mains line.

A schematic of Isolation Transformer is given below in Fig. A.

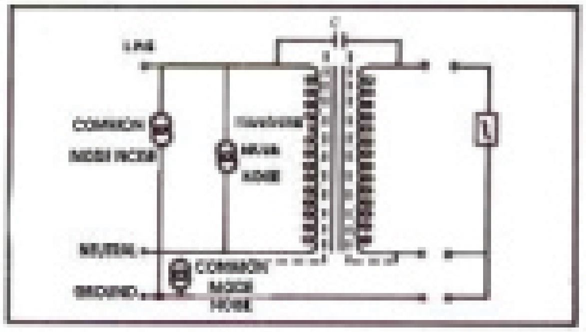

Fig. A.
Sinetrac Isolation Transformer. Two center lines represent core material C is effective coupling capacitance between primary and secondary.

"SINETRAC" Isolation Transformer, attenuates noise in both directions 'IN' as well 'OUT'. It attenuates common mode noise and transverse mode noise. Common Mode Noise occurs between the ground wire and both current carrying conductors (Fig. A) while transverse Mode Noise occurs between the current carrying conducts. (Fig. B)

**CONFIGURATION:** 'SINETRAC' Isolation Transformer employs a unique multiple shielding technique that reduces the interwinding capacitance to below 0.005 picofarads, D.C. Isolation to over 1000 Megohms and line leakage current to a safe 10 microamps maximum. Whereas an ordinary Isolation Transformer has interwinding very high capacitance. 'SINETRAC' Isolation Transformer has therefore only ten thousandth the noise leakage as compared to an ordinary Isolation Transformer. Sinetrac Isolation Transformers are available upto designed rating in single phase. For three phase Input supply, three single phase units are connected in STAR configuration.

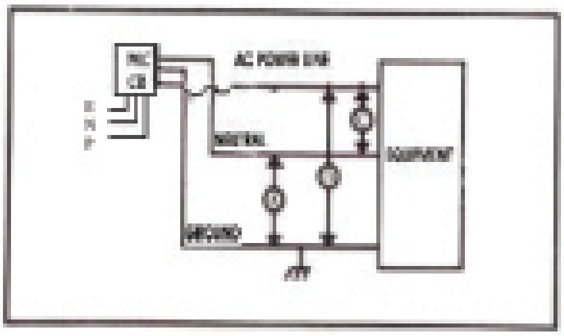

Fig B.
Noise attenuation Common mode noise (A and B) & transverse mode noise (C

## TECHNICAL SPECIFICATIONS OF SINETRAC ISOLATION TRANSFORMER

| | |
|---|---|
| Input Voltage Single Phase / Three Phase | 230 / 415 ± 10% AC |
| Mode of operation | As Step Down or Step-UP or 1:1 Isolation Transformer |
| Common Mode Noise Rejection: | Over 100 Db |
| Coupling Capacitance | Less than .1 pF |
| Insulation Level | More than 100 Meg ohms |
| Breakdown Strength | 2500 V AC for 2 minutes |
| Load Regulation | About 3.5% |
| Line Leakage Current | Less than 50 micro amps upto 1 kVA |

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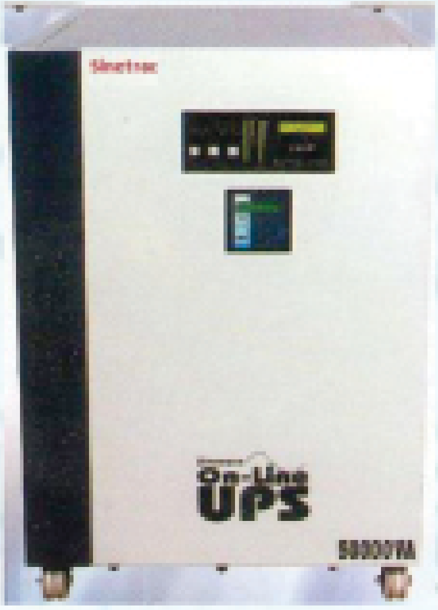
7.5kVA 1Ø In & Out



50kVA 3Ø In & Out

Sinetrac®

# ONLINE UPS

# Capacity : 3kVA to 100kVA

Sinetrac®

# Technical Specifications

| | 1Ø INPUT 1Ø OUTPUT | 3Ø INPUT 1Ø OUTPUT | 3Ø INPUT 3Ø OUTPUT |
|---|---|---|---|

**INPUT**

| | | | |
|---|---|---|---|
| Voltage | 230V – 20% + 15% | 415V -20% + 15% | 415V -20% + 15% |
| Frequency | 50Hz ± 10% | 50Hz ± 10% | 50Hz ± 10% |
| DC Voltage | 120/180V DC | 360V DC | 360V DC |

**OUTPUT**

| | | | |
|---|---|---|---|
| Voltage | 220 – 230V ± 1% | 220 – 230V ± 1% | 415V ± 1% |
| Frequency | 50Hz ± 0.01% | 50Hz ± 0.01% | 50Hz ± 0.01% |
| Power Factor | 0.7 Lag to Unity | 0.7 Lag to Unity | 0.7 Lag to Unity |
| Crest Factor | 5:1 | 5:1 | 5:1 |
| Transient Response | Output RemainsWithin5% | Output Remains Within 5% | Output Remains Within 5% |
| On 10% To 100% | Recovery Within 1 cycle | Recovery Within 1 cycle | Recovery Within 1 cycle |
| Output Isolation | In Built | In Built | In Built |
| Over Load a. | 150% For 1 Sec | 150% For 1 Sec | 150% For 1 Sec |
| b. | 110% For 1 Minute | 110% For 1 Minute | 110% For 1 Minute |
| Input Over Voltage | Rectifier Trip | Rectifier Trip | Rectifier Trip |
| Dc Under/Over Voltage | Inverter Trip | Inverter Trip | Inverter Trip |
| O/P Under/Over Voltage | Inverter Trip | Inverter Trip | Inverter Trip |
| RIF/ Emi Protection | Yes | Yes | Yes |
| Computer Interface | Through Db 9 Port (Optional) | Through Db 9 Port (Optional) | Through Db 9 Port (Optional) |
| Generator Compatible | Yes | Yes | Yes |
| Acoustic Noise | < 42 Db at one Meter | < 50 Db at one Meter | <60 Db at one Meter |
| Static By Pass | Optional | Optional | Optional |

**PROTECTIONS**

| | | | |
|---|---|---|---|
| Rectifier | Input Over Voltage/Under Voltage | Input Over &Under Voltage | Input Over & Under Voltage |
| | DC Over Voltage | DC Over Voltage | DC Over Voltage |
| | Battery Charging<br>Over Current | Battery Charging<br>Over Current | Battery Charging<br>Over Current |
| | | Single Phasing | Single Phasing |
| | | Reverse Phase Sequence | Reverse Phase Sequence |

**PROTECTIONS**

| | | | |
|---|---|---|---|
| Inverter | Output Over Voltage/Under Voltage | Output Over Voltage/Under Voltage | Output Over Voltage/Under Voltage |
| | Output Overload | Output Overload | Output Overload |
| | Output Short Circuit | Output Short Circuit | Output Short Circuit |
| | DC Under Voltage | DC Under Voltage | DC Under Voltage |
| | Over Temperature | Over Temperature | Over Temperature |

**AUDIO ALARM**

| | | | |
|---|---|---|---|
| Rectifier Trip | Mains Fail | Mains Fail | Mains Fail |
| Battery Low Pre-alarm | System Trip | System Trip | System Trip |
| Overload | System Trip | System Trip | System Trip |

**ENVIRONMENT**

| | | | |
|---|---|---|---|
| Operating Temperature | 0 to 45°C / 50°C optional | 0 to 45°C / 50°C optional | 0 to 45°C / 50°C optional |
| Storage | -10 to 70°C | -10 to 70°C | -10 to 70°C |
| Humidity | Up to 95%RH(Non-condensing) | Up to 95% RH(Non-condensing) | Up to 95% RH(Non-condensing) |
| Cooling | Forced air cooling | Forced air cooling | Forced air cooling |

**Sinetrac Other Products:**

**Servo & Automatic Voltage Stabilizers, Isolation Transformers, Online UPS, Solar Advanced Hybrid UPS, Auto Phase Correctors/Protectors, Water Level Indicators, Street & Lighting Timers, Battery Chargers, Rectifiers, Utility Gadgets etc.**

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# STREET / LIGHTING TIMERS

Sinetrac®


Conserving Energy has always been of great importance and different ways and means are being found to achieve that.

**Street Lights Dual Automation System (SLDAS)** is one of the most widely accepted products for the same. It not only helps us save energy but it also helps in precise switching 'ON' & 'OFF' street lights without manpower involvement.

**Street Lights Dual Automation System (SLDAS)** switches 'ON' the street lights on sun set when the light level dips to 40 Lux and switches 'Off' the same on day break when light level reaches 80 Lux without involving people to do the same. It too has a second option of programming the switching 'ON' & 'OFF' (with an accuracy of ±2 milli seconds) the street lights for the entire year (365 days) with four different settings available for the same.

There is a Liquid Crystal Display (LCD) displaying the date, time, running status of 'ON' or 'OFF' on the front panel with switches to pre-programme the desired timings for the entire year on Day to Day basis.

A switch has been provided for different operational modes, i.e. Light / Photo sensing Base, Pre-programmed base option and a By-pass facility to bring the system to manual mode operations.

The SLDA system not only brings the street lights on Automation it also protects them from High & Low Voltages, Overload, Short Circuit, Inbalance in Phase Voltages due to neutral break.

The system is available in Single Phase & Three Phase options. Three Phase models have another added feature of switching "OFF" the IInd & IIIrd Phase tube lights after the desired 'ON' time resulting in further energy conservation controls for setting the 'ON' time for the Y & B Phases can be used for the purpose if required.

The Street Lights Dual Automation Systems are available in Single & Three Phase ratings.

| MODELS | CAT NO. | RATING | PHASES |
|---|---|---|---|
| 1 | SLDA 10000 SP | 10 kVA | Single |
| 2 | SLDA 16000 SP | 16 kVA | Single |
| 3 | SLDA 25000 TP | 25 kVA | Three |
| 4 | SLDA 40000 TP | 40 kVA | Three |

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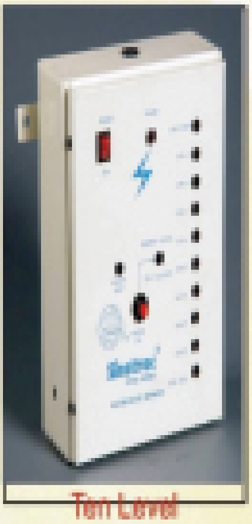
# WATER LEVEL SENSORS



Ten Level Water Indicator

## Introduction

Sinetrac

**Sinetrac Water Level Sensor** unit plays an important role in power saving as well as safety of electrical equipments connected with pumping sets.This is an IC controlled electronic type water level indicator, which indicates the percentage of water level in the overhead Tank/Sump in desired steps. The number of steps can be adjusted according to specific need, along with the visual indications. It gives alarm on low and high levels, Both visual and audio alarms are provided on low and high levels distinguishable by still light / continuous alarm in low condition and blinking light / intermittent alarm in high condition. Hence precise & instant status is available of overhead Tank/Sump. The sensors consist of a sturdy, waterproof steel housing to prevent corrosion, well gripped in a mould.

## Features

- Power Saver.
- Water sensor.
- Equipment life longevity.
- Correct & instant water level status of overhead Tank.
- Pre-warning Audio as well as visual alarms.
- Dry Run Pre-Warning.
- Pre-warning against unnecessary long run of motors & Overflow.
- By pass facility of audio alarm in case of high & low level with Visual indications to display.

## Alarms/Indications

| | Status | Audio | Visual |
|---|---|---|---|
| a) | High Level | Intermittent Sound | Blinking light |
| b) | Low Level | Continuous Sound | Still light |

## Installation

Corrosion free sensors are dipped in water at different levels mounted on a Stainless steel channel. The sensors are connected to the main equipment in the pump House with sturdy Cable to give precise & instant water level status in overhead tanks/Sump.

| Sensors | Indicators | Models | Cable |
|---|---|---|---|
| Six | 5 Levels | WLS : 5 | Six Core |
| Nine | 8 Levels | WLS : 8 | Twelve Core |
| Eleven | 10 Levels | WLS : 10 | Twelve Core |

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# Three Phase Line Protector For Protection & Automation of Pump Houses

Sinetrac®

## Introduction

The equipment provides only proper three-phase supply and acts as a safety gadget to the loads connected through it from the erratic voltages, the phase sequence, single phasing over loading etc.

## Installation

For good performance of any equipment, correct installation is perhaps one of the most important one. Hence correct positioning is essential, it should be installed as near to the Mains Supply Board as possible to maintain short cable length, neat & clean cabling.

## Lay-Out

i) Switch off the Mains Switch.

ii) Now Connect with proper rating input cables Red, Yellow, Blue and Black of the Mains Box and input Terminals to the Three Phase Protector marked by R, Y, B, N respectively.

iii) After this Connect output with a proper rating cables Red, Yellow, Blue, Black the load to be connected through the Three Phase Line Protector and output Terminals, marked by R, Y, B, N respectively.

iv) After connections the Main Switch may be switched on at the desired time for one time initial start.

## Operating Instructions

**R Y B Pilot Lamps :** Indicate the presence of three phases in input supply.

**Healthy Mains :** It states the presence of proper three phase supply i.e.

i) Normal within range mains Voltages in all three phases..

ii) Proper sequence of all the three phases.

**Overload Indication :** Provides the overload tripping facility. Over load indicator indicates that the equipment is tripped due to over loading.

**Over load Reset :** Once the equipment gets tripped, the load should be reduced and then reset. Pushbutton should be pressed to release the overload-tripped condition.

**Timer On :** The Three Phase Line Protector has the optional facility to work all by itself. So it can be installed at un-manned stations. As soon as the timer is switched on it starts giving the output for desired hours to the equipment and then switches off for desired time the output supply automatically, then again it restarts and remains

'ON'& 'OFF' automatically. This continues to work perfectly till the circuits are being fed voltage through mains or Battery. In case the time has to be rescheduled the system should be switched off by start switch in 'UP' position and again brought down at "the time of start".

**The timer 'ON' & 'OFF' cycles can be adjusted on site or preset from factory as per order.**

# Available Models

A) TPP | B) TPPAC | C) TBES | D) ACES

## A Model : TPP Features

| | | |
|---|---|---|
| (i) Single Phasing protection(SPP) | — | No Output (Trips) |
| (ii) High/ Low Voltage Protection(HLVC) | — | No Output (Trips) |
| (iii) Phase Reversal Protection(PR) | — | No Output (Trips |
| (iv) Over Load Protection | — | No Output (Trips) |

**Available Capacities**

| Models | Copper Cables | Models | Copper Cables |
|---|---|---|---|
| 10kVA Loads upto 11hp, | 4 sq. mm | 60kVA Loads upto 67hp, | 35 sq. mm |
| 15kVA Loads upto 18hp, | 6 sq. mm | 75kVA Loads upto 85hp, | 50 sq. mm |
| 25kVA Loads upto 28hp, | 16 sq. mm | 100kVA Loads upto 125hp, | 70 sq. mm |
| 40kVA Loads upto 45hp, | 25 sq. mm | 135kVA Loads upto 150hp, | 120 sq. mm |
| 50kVA Loads upto 56hp, | 35 sq. mm | 155kVA Loads upto 173hp, | 150 sq. mm |

## B Model : TPPAC Features

| | | |
|---|---|---|
| (i) Single Phasing protection(SPP) | — | No Output (Trips) |
| (ii) High/ Low Voltage Protection(HLVC) | — | No Output (Trips) |
| (iii) Phase Reversal Protection(PR) | — | Corrected output of Phases in sequence available |
| (iv) Over Load Protection | — | No Output (Trips) |

**Available Capacities**

| Models | Copper Cables | Models | Copper Cables |
|---|---|---|---|
| 10kVA Loads upto 11hp, | 4 sq. mm | 60kVA Loads upto 67hp, | 35 sq. mm |
| 15kVA Loads upto 18hp, | 6 sq. mm | 75kVA Loads upto 85hp, | 50 sq. mm |
| 25kVA Loads upto 28hp, | 16 sq. mm | 100kVA Loads upto 125hp, | 70 sq. mm |
| 40kVA Loads upto 45hp, | 25 sq. mm | 135kVA Loads upto 150hp, | 120 sq. mm |
| 50kVA Loads upto 56hp, | 35 sq. mm | 155kVA Loads upto 173hp, | 150 sq. mm |

## C Model : TBES Features

| | | |
|---|---|---|
| (i) Single Phasing protection(SPP) | — | No Output (Trips) |
| (ii) High/ Low Voltage Protection(HLVC) | — | No Output (Trips) |
| (iii) Phase Reversal Protection(PR) | — | No Output (Trips) |
| (iv) Over Load Protection | — | No Output (Trips) |
| (v) Different Setable times facility for | — | Connected load gets a automatic start and off in accordance with their settings On time & OFF time |
| (vi) Built in Electronic Star Delta Starter Facility | — | Automatic Star Delta conversion in every cycle |

**Available Capacities**

| Models | Copper Cables | Models | Copper Cables |
|---|---|---|---|
| 10kVA Loads upto 11hp, | 4 sq. mm | 60kVA Loads upto 67hp, | 35 sq. mm |
| 15kVA Loads upto 18hp, | 6 sq. mm | 75kVA Loads upto 85hp, | 50 sq. mm |
| 25kVA Loads upto 28hp, | 16 sq. mm | 100kVA Loads upto 125hp, | 70 sq. mm |
| 40kVA Loads upto 45hp, | 25 sq. mm | 135kVA Loads upto 150hp, | 120 sq. mm |
| 50kVA Loads upto 56hp, | 35 sq. mm | 155kVA Loads upto 173hp, | 150 sq. mm |

## D Model : ACES Features

| | | |
|---|---|---|
| (i) Single Phasing protection(SPP) | — | No Output (Trips) |
| (ii) High/ Low Voltage Protection(HLVC) | — | No Output (Trips) |
| (iii) Phase Reversal Protection(PR) | — | Corrected output of Phases in sequence available |
| (iv) Over Load Protection | — | No Output (Trips) |
| (v) Different Setable times facility for On time & OFF time | — | Connected load gets a automatic start and off in accordance with their settings |
| (vi) Built in Electronic Star Delta Starter Facility | — | Automatic Star Delta conversion in every cycle |

**Available Capacities**

| Models | Copper Cables | Models | Copper Cables |
|---|---|---|---|
| 10kVA Loads upto 11hp, | 4 sq. mm | 60kVA Loads upto 67hp, | 35 sq. mm |
| 15kVA Loads upto 18hp, | 6 sq. mm | 75kVA Loads upto 85hp, | 50 sq. mm |
| 25kVA Loads upto 28hp, | 16 sq. mm | 100kVA Loads upto 125hp, | 70 sq. mm |
| 40kVA Loads upto 45hp, | 25 sq. mm | 135kVA Loads upto 150hp, | 120 sq. mm |
| 50kVA Loads upto 56hp, | 35 sq. mm | 155kVA Loads upto 173hp, | 150 sq. mm |

**Abbreviations :**

**SPP** : Single Phase Protection ; **TPP** : Three Phase Protector ; **TB** : Timer Based ; **HLVC** : High & Low Voltage Cut ; **PR** : Phase reversal; **AC** : Auto Corrector ; **TPPAC** : Three phase protector Auto Corrector ; **TBES** : Timer Base Electronic Starter ; **ACES** Auto Corrector Electronic Starter ; **DR** : Dry Run Protection

**Note:**

(i) Models TBES -10kVA, 15kVA & ACES 10kVA, 15kVA will have DOL starter. Remaining all TBES and ACES Models will have star-delta starters.

(ii) Dry run Protection available with select models and is optional on an extra charge.

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Website: www.sinetrac.in

# Battery Chargers CCCV Type

Battery Charger 12-72BC20A

## Introduction

**Sinetrac Battery Chargers** are SCR controlled and essentially consist of a double wound transformer, an SCR/Diode bridge, smoothing choke, capacitors and electronic circuits for controlling the voltage and current. All these components are housed in a sheathed steel cabinet suitable for floor mounting and fitted with a Voltmeter, an Ammeter, fuses etc. The transformers and coils are double wound and contain lamination worldwide acknowledged for good efficiency.

## Principle of Operation :

The AC supply from the Mains is applied to the primary of main transformer to obtain the required voltage. This is followed by a bridge circuit comprising of two diodes and two silicon controlled rectifiers (SCR). Surge suppression circuits made up of resistances and capacitors are used to protect the power device from the transients and surges. The output of the bridge is connected to conventional LC filter network. The output of this circuit depends upon the firing angle of the SCRs. There are two basic modes of operation.

- Constant Voltage ( Float mode)
- Constant Current (Boost mode)

## Constant Voltage Mode :

The voltage across the output is sensed by the feedback and is compared with a comparator. The error voltage controls the firing angle of the SCR to maintain a constant voltage. Output voltage can be adjusted in this mode from 2V to 2.3 V /Cell with the help of Trimpot.

## Constant Current Mode :

The output current is sensed by a shunt. The voltage developed across this shunt is again compared with a constant voltage reference, which is used to properly fire the SCRs to maintain a constant current. In this mode current can be adjusted by a current control Trimpot and relatively the load voltage changes from 1.7v to 2.4v /cell.

To start with the charger functions in float mode. However if the battery charging current requirement is more than 1/10th of the boost charging current then the charger will automatically switch over to the boost mode for charging. It shall continue under Boost mode till the battery voltage reaches 2.4v /cell and consequently charging current comes down to 1/10th of the boost charging current.

## Readily Available Models

| Models | Battery Charging Capacity | Models | Battery Charging Capacity |
|---|---|---|---|
| 12-24 BC10A | 1 No. or 2 No's 12 volts upto 100Ah | 12-24 BC10A | 1 No. or 2 No's 12 volts upto 200Ah |
| 12-48 BC10A | 1, 2, 3 or 4 No's 12 volts upto 100Ah | 12-48 BC10A | 1, 2, 3 or 4 No's 12 volts upto 200Ah |
| 12-72 BC10A | 1, 2, 4 or 6 No's 12 volts upto 100Ah | 12-72 BC10A | 1, 2, 4 or 6 No's 12 volts upto 200Ah |

## Indications :

i. Mains supply

ii. Mode of charging (Float or Boost)

iii. Polarity Battery Reverse

iv. Over load condition

## Protections

i) Over load protection
ii) Short circuit protection
iii) Reverse Polarity protection
iv) Surges & Transient protection

## Installation & Commissioning

The Battery charger should be installed at a place where the ambient temperature doesn't exceed 45° C. After opening the packing case, it should be ensured that no physical damage has been caused to the battery charger during transportation. Avoid excessive corrosive fumes and ensure free air circulation. Adequate space to be provided for effective ventilation. Before plugging the charger to AC Mains, check all fuses, meters and if possible the insulation resistance using a megger.

## Operating Instruction

- Before connecting the charger to the AC Mains, make a thorough physical check for any missing/broken Fuse and Meter etc.
- Connect the input Lead of charger to an AC Mains socket of rated input current of the charger while the power supply voltage should range between 180-270V. Input AC Mains indicator will glow. Though the charger will still charge but not all that efficiently on lower voltage.
- Connect the battery(ies) to be charged to the output terminals of the charger with correct polarities and commence charging.
- If the batteries are not connected properly to the output terminals of the charger, MCB will trip and indication of reverse polarity will glow.
- If there is any short circuit on the output side of the charger in this condition MCB will also trip and charging will not commence.

## Technical Specification :

| | | |
|---|---|---|
| Input working voltage | – | 180-270 Volts |
| Nominal Input Voltage | – | 230V AC, 50Hz, 1 |
| On load input Power Factor | – | >0.8 |
| On load unit watt efficiencies | – | >80% |
| No load losses of complete charge | – | < 10% |
| Ripple contents on the load (rms) | – | 1% to 5% |
| Over load | – | 110% for 30 minutes and will give visual indication for over load condition. |
| Boost Charging | – | 2.4 V/cell |
| Float charging | – | 2.15 V/cell |
| Float cum Boost Charging | – | Automatically changes from one state to another state as per battery condition and will give the visual indication for either float or Boost based on charging mode. |
| Reverse Polarity | – | No damage will be caused. A visual indication for Reverse polarity will appear with an audio alarm |
| Short circuit | – | MCB protection provided. |

*Due to continuous upgradation specifications may change without prior notice*

**Chime Electronics**

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Website: www.sinetrac.in


Sinetrac®

Pure Sine Wave

Advanced Hybrid

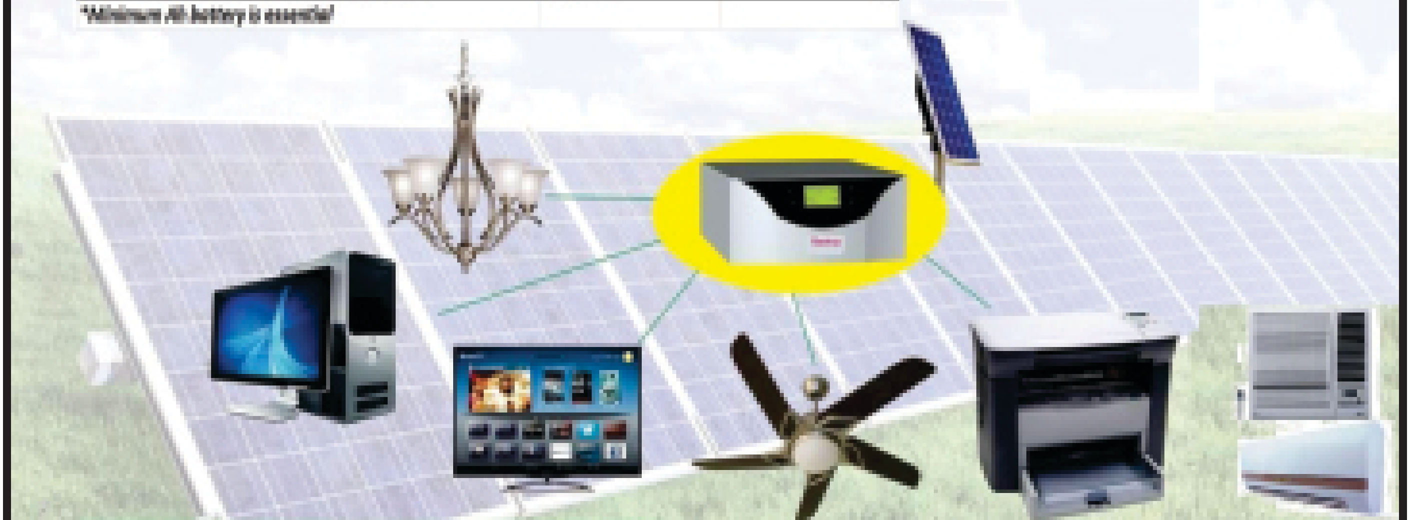
# Solar UPS

CHIME ELECTRONICS
DELHI (INDIA)

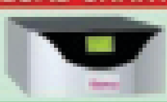
Technical Specifications*

| Parameters | Normal Mode | UPS Mode |
|---|---|---|
| Mains AC Voltage Range | 100-280 V (Wide) | 180 – 260 V |
| Output Voltage | 230V ± 10V | 230V ± 10V |
| Output Frequency | 50 Hz ± 0.5 Hz | 50 Hz ± 0.5 Hz |
| Battery Charging Volt. Range | 100 to 280 V | 180 to 260 V |
| Change Over Time | <40 mSec | <10 mSec |
| Battery Low Cut Voltage | 10.5 Volt | |
| Charging Current (Mains) | Normal Charge - 10A ± 1A<br>High Charge - 12A ± 1A | |
| Output Waveform | Pure Sine Wave | |
| Normal/UPS Overload | 1.2 | |
| Normal/UPS Short Circuit | 3 | |
| Technology | DSP based Sine Wave Technology | |
| Auto Reset Feature | Available | |
| | 1.45 KVA UPS - 24V PV Panel<br>2 KVA UPS - 24V PV Panel<br>2.5 KVA UPS - 48V PV Panel<br>3.5 KVA UPS - 48V PV Panel<br>5 KVA UPS - 96V PV Panel<br>7.5 KVA UPS -120V PV Panel<br>10 KVA UPS - 120V PV Panel<br>12 KVA UPS - 192V PV Panel<br>15 KVA UPS - 240V PV Panel | |
| Hybrid Solar Charger Controller | | |
| Technology | DSP based intelligent battery charging and Charge Sharing with mains | |
| Charge Controller Type | PWM based | |
| Solar PV Current | 40 Amps Lower KVA<br>75 Amps Higher KVA | |
| Peak Battery Charging Current (Solar) | 30 Amps | |
| Charge Controller Efficiency | >95% | |
| PV Reverse Plarity Protection | Provided | |
| Reverse Current flow to PV module Protection | Provided | |

*Minimum Ah battery is essential



LOAD CHART



| OPTIONS | A | B | C | D |
|---|---|---|---|---|
| COMPUTER | | | | 1 |
| PRINTER | | | | 1 |
| TELEVISION | | | 1 | |
| TUBE LIGHT | 8 | | 6 | 4 |
| FAN | 8 | | 6 | 4 |
| CFL | 4 | 62 | 9 | |

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